PUSHFORWARDS SUPPORTED ON A SECTION: UNIQUENESS OF THE SECTION WHEN Y IS REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0FZP: sources/stacks-chow/tags.txt, source lines 19959–19959. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a finite type separated morphism of schemes with a section $s : Y \to X$. Let $\mathcal{F}$ be a finite type quasi-coherent module on X, set theoretically supported on $s(Y)$ with $\mathcal{L} = f_*\mathcal{F}$ an invertible $\mathcal{O}_X$-module. If Y is reduced, then $\mathcal{F} \cong s_*\mathcal{L}$.*

Proof. By Lemma 0FZP there exists a section $s' : Y \to X$ such that $\mathcal{F} = s'_*\mathcal{L}$. Since $s'(Y)$ and $s(Y)$ have the same underlying closed subset and since both are reduced closed subschemes of X, they have to be equal. Hence $s = s'$ and the lemma holds. □

Date: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.